

A seminorm with two-sided bounds against a reference norm

MathlibAnnex.EquivalentSeminorm

structure

Records a norm through a seminorm, two positive comparison constants and continuity on the original normed space.

Statement

Let X be a real normed space with reference norm $\|\cdot\|$. An equivalent-seminorm datum consists of a real seminorm p and constants $c, C > 0$ such that

$$c\|x\| \leq p(x) \leq C\|x\| \quad (x \in X),$$

together with continuity of p for the reference topology. The record keeps this comparison data explicit.

Definition

The eight fields are: (1) the seminorm p ; (2) a real lower constant c ; (3) a real upper constant C ; (4) a proof that $c > 0$; (5) a proof that $C > 0$; (6) the lower comparison $c\|x\| \leq p(x)$ for every x ; (7) the upper comparison $p(x) \leq C\|x\|$ for every x ; and (8) continuity of p . These fields specify data and its properties; constructing such a record requires supplying all eight.

Assumptions

The carrier X is a real normed vector space. The stored function $p : X \rightarrow \mathbb{R}$ is a seminorm: it is subadditive, nonnegative and satisfies $p(tx) = |t|p(x)$. Both comparison constants are strictly positive, both inequalities hold for every x , and continuity of p is part of the stored data. There is no finite-dimensionality or completeness assumption.

Conclusion

The lower bound makes p definite: if $p(x) = 0$, then $0 \leq c\|x\| \leq 0$; since $c > 0$, $\|x\| = 0$ and $x = 0$. This is the separately proved definiteness lemma. The related definitions give $B_p = \{x : p(x) \leq 1\}$ and $S_p = \{x : p(x) = 1\}$. In particular $B_p \subseteq \{x : \|x\| \leq 1/c\}$, by the bounded-ball lemma.

The reference norm on X remains in place. The separate construction $\text{Space}(p)$ equips a copy of the carrier with norm p ; its distance is $p(x - y)$, as recorded in the cited transport theorem. Boundedness of B_p alone does not assert compactness in an arbitrary infinite-dimensional space.

Main citations

- Exact declaration and its source — `MathlibAnnex.EquivalentSeminorm`
- Closed seminorm unit ball — `MathlibAnnex.EquivalentSeminorm.closedUnitBall`
- Unit seminorm sphere — `MathlibAnnex.EquivalentSeminorm.unitSphere`
- Membership in the closed ball — `MathlibAnnex.EquivalentSeminorm.mem_closedUnitBall`
- Definiteness from the positive lower bound —
`MathlibAnnex.EquivalentSeminorm.eq_zero_of_apply_eq_zero`
- Boundedness in the reference norm —
`MathlibAnnex.EquivalentSeminorm.isBounded_closedUnitBall`
- Distance on the separately normed carrier — `MathlibAnnex.EquivalentSeminorm.dist_space_eq`

Lean source signature (exact)

```
structure EquivalentSeminorm (E : Type*) [NormedAddCommGroup E] [NormedSpace ℝ E] where
  p : Seminorm ℝ E
  lower : ℝ
  upper : ℝ
  lower_pos : 0 < lower
  upper_pos : 0 < upper
  lower_le : ∀ x, lower * ‖x‖ ≤ p x
  le_upper : ∀ x, p x ≤ upper * ‖x‖
  continuous_p : Continuous p
```

In the source	Mathematical meaning
<code>(E : Type*) [NormedAddCommGroup E] [NormedSpace ℝ E]</code>	The real normed space X , with its reference norm $\ \cdot \ $.
<code>p : Seminorm ℝ E</code>	The function $p : X \rightarrow \mathbb{R}$, nonnegative, subadditive and absolutely homogeneous: $p(tx) = t p(x)$. Definiteness is not part of the seminorm type.
<code>lower : ℝ</code>	The stored lower comparison constant c .
<code>upper : ℝ</code>	The stored upper comparison constant C .
<code>lower_pos : 0 < lower</code>	The requirement $c > 0$.
<code>upper_pos : 0 < upper</code>	The requirement $C > 0$.
<code>lower_le : ∀ x, lower * ‖x‖ ≤ p x</code>	For every $x \in X$, $c\ x\ \leq p(x)$.
In the source	Mathematical meaning
<code>le_upper : ∀ x, p x ≤ upper * ‖x‖</code>	For every $x \in X$, $p(x) \leq C\ x\ $.

In the source	Mathematical meaning
<code>continuous_p</code> : Continuous p	Continuity of p in the reference norm topology. Together with the preceding seven rows this gives all eight stored fields.

▼ Exact surrounding binder context (separate excerpt)

```
namespace MathlibAnnex
```

▼ Exact content identity

Declaration: `MathlibAnnex.EquivalentSeminorm`

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